

A Preventive Environment for Pest Over the Crops using Simplified Network Architecture

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Abstract- Crop pests can significantly impact productivity, making early detection and prevention crucial for management. Machine learning approaches are being used to forecast crop diseases using a variety of data sets. Relatively easy access to the environment data has resulted in a lot of attempts at predicting diseases and pests using this data. In this research, we provide a model that uses deep learning technique known as Simplified YOLO 3.1 architecture to examine past crop growth environment data to forecast illnesses. Utilizing large public datasets on crops such as tomato, pepper, grape, strawberry, and paprika, we show that the model is capable of accurately forecasting crop pest and disease risk. With an average accuracy of 98% and precision of 96%, it showed excellent predictive power. The anticipated results can help with post-harvest processing or insect control. It is expected that this learning framework and crop disease prediction model, which is based on environmental data, will be extensively useful for the prevention of pests and diseases in various facilities and crops.

Keywords- pest, control, prediction, infection, learning model

1. Introduction

Crop diseases and pests raise production costs by diminishing both the yield and quality of crops, as well as increasing labor requirements. They also lead to environmental harm due to the excessive use of chemicals for pest control [1]. These issues are significant threats to food security, contributing to a global grain yield loss of approximately 15–40%. Specifically, a serious disease that affects crops can cause output losses of up to 70% that is the subject of this research

[2]. Furthermore, another dangerous disease, gray mold, usually destroys 20% of strawberries, with losses of more than 60% in extreme cases [3]. Trillions of US dollars are spent on the pesticides used to address these problems, but only 0.1% of them really kill the pests; Pollution results from the absorption of the remainder and their discharge into the environment. Predicting the behavior of pests has been the focus of several attempts to lessen the damage they bring to the economy and environment. These disease modeling initiatives involve creating mathematical models that describe how different factors influence the emergence and growth of pests. These models are utilized to predict the possibility and intensity of pest outbreaks, comprehend the diseases and pests that diminish crop yield and quality, and raise costs due to the need for more labor in pest management and disease development. Additionally, they aid in making tactical decisions [4]. Crop growth variables, such as humidity, temperature, and rainfall, are important factors in pest modeling. The crop's growing environment is a key factor in the emergence of pests and diseases and it is frequently included in modeling due to the fact that, in contrast to other parameters like host nutrients, host genetics, and microbial pathogens, it is comparatively simple to gather and handle data about [5] – [6]. For instance, the model executes a criterion that identifies infections by computing the amount of intermediate variables based on environmental criteria like air temperature, rainfall, leaf wetness, and relative humidity,

The evaluation is employed to propose that pests is present or not. By calculating a comprehensive late blight severity index that considers thermal and humidity appropriateness based on

temperature and humidity data, the potato late blight forecasting model also assesses pest occurrence. As the [7] model illustrates, these conceptual modeling techniques enable the interpretation of predictions, including the biological systems at play. But to predict pests and diseases, these models depend on a lot of variables, and in order to integrate these variables efficiently, a thorough knowledge of the target crops and pests is necessary. As a result, if the model is applied to a different target crop, pest, or disease, a new model may be required, or considerable alterations to existing ones. It is not feasible to create models for several kinds of pests. The data-driven strategy is extensively investigated due to its capacity to use huge and complicated datasets, thanks to advancements in big data and deep learning technologies. Environmental data and image data are increasingly being integrated into multi-modal formats. Since facility farms can quickly install the necessary technology for data collection, growing environment data is one of the simplest forms of data to collect on a large scale [6].

There isn't currently a model that solely utilizes crop growth environment variables, as far as we know to predict or forecast pest infestations in a universal setting using deep learning techniques. By analyzing crop growth environmental data to anticipate disease and pest infestations and computing the risk score that indicates the normal-infestation situation, we provide the deep learning-based predictive modeling technique in this study. Utilizing growth environment data gathered from several facilities, a prediction model was developed, demonstrating the methodology's universal applicability despite differences in farm infrastructure, crop types, and target pests. Here, a novel Simplified YOLO 3.1 architecture is designed to reduce the network complexity and to enhance the model performance. The model gives promising outcomes in predicting the pest over the crops. The work is extended as: section 2 explores the review and advancements of Simplified YOLO 3.1 architecture is given in section 3. The metrics are practically analyzed in section 4. The work summary is given in section 5.

2. Related works

The literature explores various facets of pest detection, examining multiple datasets and detection algorithms focused on identifying small pests and pest species exist in various kinds of environments. Author in [7] developed a system for predicting traps using photos obtained from traps in a greenhouse scenario. They created a two-stage bespoke detector that surpassed Faster R-CNN, which was noted for its poor ability to identify microscopic pests. Nevertheless, this approach relies on pictures from sticky traps, which often yield lower accuracy when compared to images with complex backgrounds. Employing deep learning and object identification techniques, [8] aimed to develop models that could recognize three types of pest moths in images taken from pheromone traps. We tested the speed and accuracy of seven distinct deep learning object detection algorithms using pheromone traps. Among them, Faster R-CNN had the highest precision.

However, the dataset employed in this study had limitations because it did not include actual field photos. The author in [9] conducted a thorough evaluation of recent advances in automatic pest detection utilizing close-range photos and machine learning approaches. They emphasized the necessity for greater pest image collections and the challenge of creating efficient pest detection systems. They suggested solutions including developing data exchange amongst research teams and participating farmers initiatives. They also examined the shortcomings of existing monitoring systems including the essential for enhanced solutions for efficient processing, expense and high-density image sensors complexity. According to the research, while automated pest monitoring offers significant challenges, advances in machine learning techniques show promise for practical applications. To automatically identify pests in traps, the author et al. [10] developed deep learning models using a CNN model to recognize and filter out irrelevant objects. This study's main objective was to create the model for calculating the overall pests in photos. To identify thrips and whiteflies in photographs taken under controlled conditions, the author set out to create a deep learning-based identification model. A hybrid

model called the RCNN was utilized to detect tiny pests using sticky traps [11].

In reference [12], a pre-trained deep CNN framework was proposed for the identification and classification of pests, achieving an accuracy of 88.83% with the DenseNet169 model. The author employed pheromone traps to detect the pest which lives on black pine. They used powerful deep learning object detection algorithms to count and identify pests in photos from these traps. However, when applied to photos collected in real-time or in more complicated scenarios, the trained algorithm may perform less well. To identify pests and illnesses in plants, the author [12] reviewed techniques used in computer vision and deep learning. In order to increase accuracy and classification, they pointed out that convolutional neural networks (CNNs) need to be customized for feature extraction.

The study also highlighted that factors such as image resolution, quality of hardware, and the size of the dataset significantly affect the training of CNNs, as well as the algorithm's accuracy and efficiency. The author developed the integrated approach known as LSTM-based CNNs which combines LSTM networks with pre-trained CNN models. This model utilized various feature extraction like GoogleNet, DenseNet201 and AlexNet. By hauling features from pest images and processing them via an LSTM layer to create the model, the research aims to identify diseases. The author in [12] introduced a convolutional network to identify and categorize agricultural pests into different types. A key element of this study was the development of a feature fusion that produces integrated maps and aids in the detection of pests at different scales, particularly microscopic ones. A more complete dataset gathered in real-time might produce better results, moreover the pictures used in this study were taken from Google. The model they proposed achieved a mAP of 57%, which could be enhanced since the pests in the images were quite small, significantly impacting the overall accuracy.

3. Methodology

3.1. Data acquisition

A large dataset was gathered to create a model that uses information about the agricultural growth environment to predict disease and pest risk and probability. This study used the 'plant disease-causing dataset' that AI-hub made available which contains information on diseases related to the environmental conditions of crops grown in various facilities. Data on cumulative environmental conditions and disease outbreaks for at least 48 hours before the observation period are included in the dataset. Temperature and humidity are at least included in the environmental data, with additional information such as CO_2 levels, solar radiation, and dew point, depending on the equipment used in the facilities. The database provided us with environmental and pest infestation information for five distinct crops, with two pest infestations documented for each crop. Along with other environmental variables unique to each site, the collected data includes a variety of environmental information, such as temperature and humidity for all datasets. For grape data, dew point measurements are included, while strawberry, pepper, and tomato data feature both dew point and CO_2 concentration. Paprika data also contains measurements of sun radiation. The range of crop kinds and the many environmental monitoring methods utilized on actual farms naturally provide a challenge with the variety of data.

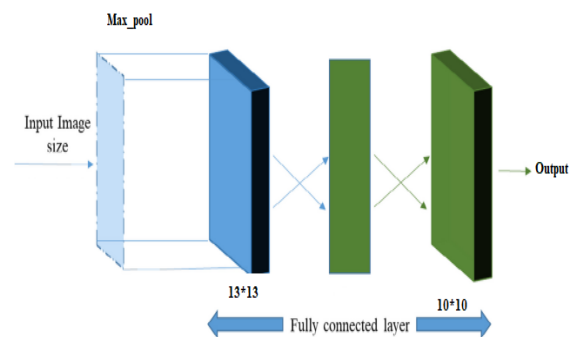


Fig 1: Simplified YOLO 3.1 architecture

3.2. Prediction model

A simplified version of YOLO V3, Simplified YOLO 3.1 architecture has a shallower convolutional layer depth than its predecessor. It extracts features by using the DenseNet

architecture. Fig 1 shows how Simplified YOLO 3.1 works. In this model, the convolutional layer is made smaller while a pooling layer is added. It first divides the image into 3×3 grid and then identifies 2D image with object scores, class predictions and bounding boxes at two different scales. The final phase of object detection involves discarding bounding boxes with the least object scores. Fig 1 illustrates the complex structure of YOLOv3-Tiny, which consists of nine convolutional layers and six layers with maximum pooling for effective feature extraction. Some bounding box is constructed model at diverse scales uses 10×10 feature map and the composite features generated by an unsampled 10×10 feature map with 13×13 features. This simple design of Simplified YOLO 3.1 allows for rapid detection speeds, but it compromises the model's accuracy in comparison to YOLOv3.

4. Numerical results

With the environmental data correspondent to crop growth, the simplified YOLO3.1 architecture is anticipated to predict if pests are available or not. According to our research, ongoing environmental data monitoring can help with early illness and pest detection and prevention. In order to do this, we collected proprietary and public data on crop conditions and growth conditions. Additionally, to manage large-scale and diverse data, we created a separate model for different variables, including types of environmental factors, measurement frequencies, and crop varieties. Using the environmental data from the preceding two days, we may use this method to forecast the state of a crop. Furthermore, from the latent space, we may calculate a risk score, which, unlike simple binary classification, illustrates a transitional phase between healthy and diseased groups. During the transition phase, it will be feasible to diagnose and predict issues before they arise. Managing the environment to reduce the risk score helps safeguard a crop that is in this transitional stage before pest and disease outbreaks occur. This research shows that it is possible to regulate the environment then use simulation to change environmental conditions while looking for

patterns in the latent space to bring the data further to a normal state.

Table 1: Performance evaluation

Model	Avg. precision	Loss	Accuracy
Unet	58%	1.4	60%
VNet	33%	0.9	45%
YOLOv3	71%	0.48	75%
DenseNet	45%	0.79	50%
Simplified YOLO 3.1 architecture	95%	0.56	98%

Beyond the environmental conditions this study looked at, a number of other factors also affect crop pests and illnesses. This study ignored a number of other factors that are known to affect crop growth, including air movement, rainfall, microbial characteristics, pH, and soil composition. Furthermore, several elements—like weather, pollinators, and weeds—are difficult to quantify for data collecting in open-farm environments. In comparison to indoor farms, which have many environmental characteristics under control, open-field farms offer a multitude of variables, making it difficult to create a model that accounts for them all. It is essential to establish the monitoring system gathers high-resolution data by evaluating aspects related to environmental factors to impact pests and diseases.

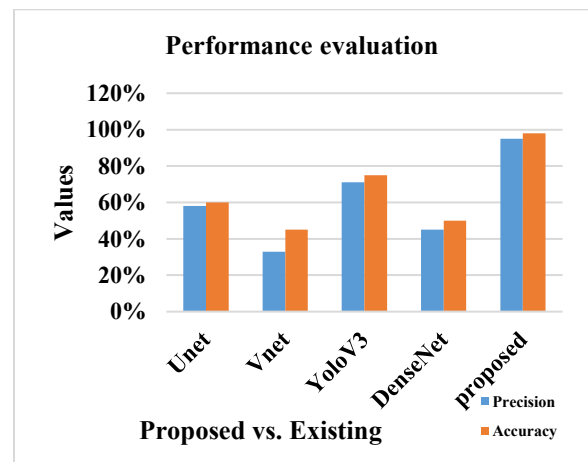


Fig 2: Performance computation

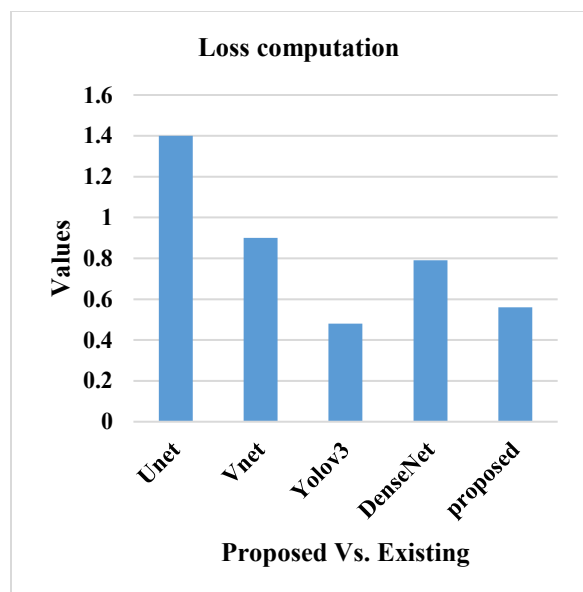


Fig 3: Loss Computation

Furthermore, more complexity is required for the model in order to take into consideration the interactions between various pests or diseases and the optimal growing conditions for crops. Resilience and resistance to challenging environments differ from optimal growth conditions, based on the stage of crop growth. With this in mind, adding information about target crops' growth stages or temporal elements to the training phase could enhance performance and practicality throughout their life cycle. However, establishing a suitable database for these tasks is essential. This model is quite adaptable since it can be adapted to different scenarios by altering variables like the environmental variable or the monitoring duration. Adding more sensors to detect different environmental parameters might increase the model's resolution, performance, and usefulness. Additionally, various time series data collected by sensors can also be integrated into this approach. In a controlled experimental setting, this model can be utilized for physiological research, including the interactions between crops and pests, as well as how crops respond to changes in their environment. Additionally, proactive pest and disease management or prevention can improve crop quality, increase yields, and reduce labor requirements in established facility farms.

5. Conclusion

To manage agricultural pests, the study presented a deep learning object identification model. It concentrated on how well Simplified YOLO 3.1 architecture identified the crop pests. Real photos were collected for crop pest control. Following model training, real-world pest photos were used to assess the algorithms' prediction results. Simplified YOLO 3.1 architecture outperformed the other models in terms of pest detection, 95% was the highest precision. The conversion to use images for real-time pest identification. It is an excellent option for real-time pest identification due to its ability to precisely identify pests of various sizes. Additionally, individuals without expertise can utilize this model for pest detection. Yet, other models, such UNet and VNet, differ in their structures and feature extraction techniques may not be as efficient and accurate at detecting small objects, even though they are still effective. In comparison to previous YOLO-based models, simplified YOLO 3.1 architecture is the most effective model for identifying even microscopic pests and integrating with technology for targeted spraying to increase accuracy and efficiency. Furthermore, the dataset gathered for this work significantly advances pest management practices and may be incorporated into other models to increase accuracy and precision. According to its conversion into the application, the Simplified YOLO 3.1 architecture methodology is now accessible in pest management as well. This work offers the significant advancement in agricultural pest management by showcasing the learning models' efficiency for pest identification and use in conjunction with real-time spraying technology.

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